

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 3.2: Linear Motion — Teacher Reference (Pre-filled)

SUMMARY TABLE: GENERAL SCIENCE GRADE 10	
Sub-Strand	Sub-Strand 3.2: Linear Motion
Driving Question	DRIVING QUESTION: Why does a runaway mkokoteni on the Limuru escarpment keep getting faster — and how can we use mathematics to predict exactly how fast and how far it will travel before it crashes?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Runaway Mkokoteni: Why Does It Keep Getting Faster?	We watched or studied a runaway mkokoteni (300 kg loaded with maize) released from rest on the Limuru-Nairobi escarpment road. It started slowly, then went much faster, and crashed into a parked lorry after 100 m in about 10 s. The gaps between its positions grew each second, and its average speed was 10 m/s.	Distance is total path length (scalar, m); displacement is the straight-line change in position with direction (vector, m). Speed is distance/time (scalar); velocity is displacement/time (vector). Acceleration is change in velocity per unit time (m/s^2). A 400 m lap has distance 400 m but displacement 0 m. The cart's velocity rises by 2 m/s every second, so it accelerates at 2 m/s^2 .	The cart has no engine and no one pushing it, yet it keeps getting faster because gravity pulls it down the slope all the time, so its velocity increases steadily from zero. It starts slowly because its velocity begins at zero. We still need mathematics (the equations of linear motion) to predict exactly how fast and how far it travels, which we will build in the next lessons.
Lesson 2 Equations of Linear Motion: Choosing the Right Equation	What did we observe or work out today? (Examples: the matatu on Thika Road reaches $v = 12 \text{ m/s}$ and $s = 48 \text{ m}$ after 8 s; the mkokoteni table for $t = 1$ to 5 s gives $v = 2, 4, 6, 8, 10 \text{ m/s}$ and $s = 1, 4, 9, 16, 25 \text{ m}$.)	What did we learn about the equations of linear motion? (Include the five variables u, v, a, s, t with units, the four equations, and the rule for choosing: list the knowns and the unknown, then pick the equation that has exactly those four variables.)	How does this help explain the driving question? (Use the cart's constant acceleration of 2 m/s^2 to explain why the velocity rises by 2 m/s every second, why the cart felt sudden, and how the equations let us predict its speed and distance at any moment.)
Lesson 3 Problem Solving with the Equations of Linear Motion: Slowing Down and Going Upward	We solved the Problem Set. The Rift Valley runner reached 10 m/s after 2.5 s and covered 12.5 m. The boda-boda travelling at 15 m/s with $a = -3 \text{ m/s}^2$ stopped after 5 s and 37.5 m. The ball thrown upward at 15 m/s from the 20 m Mombasa rooftop rose 11.25 m more in 1.5 s (31.25 m above the	We learned to pick a positive direction first, list the knowns with signs, and choose the equation that has the unknown. Slowing down gives a negative a (opposite sign to v). For a ball thrown upward, $a = -10 \text{ m/s}^2$ all the time, even at the top, where $v = 0$. The same equations work for speeding up,	The mkokoteni keeps getting faster because its acceleration is in the same direction as its motion ($a = +2.0 \text{ m/s}^2$ down the slope), so its velocity grows by 2 m/s every second. A braking boda-boda has a negative acceleration because it opposes the motion, so its velocity falls. Because

	ground), reached the ground at $t = 4$ s and hit it at -25 m/s (25 m/s downward).	slowing down and vertical motion, just like for Mzee Kamau's mkokoteni ($a = +2.0$ m/s ² down the slope).	acceleration is constant in each case, the equations of linear motion let us predict speed, distance and time at any moment, and a consistent sign convention is what makes these predictions correct.
Lesson 4 Investigating Free Fall: Predict and Observe	We dropped a rubber bung, a stone and a crumpled paper ball from 2.0 m. Mean fall times were 0.63 s, 0.64 s and 0.65 s (class mean 0.64 s), almost the same although the masses were very different. A flat sheet of paper fell more slowly than the crumpled ball.	When air resistance is negligible, all objects fall with the same acceleration due to gravity, regardless of mass. Using $s = \frac{1}{2}at^2$ with $u = 0$, $g = 2h/t^2 = 4/0.64^2 = 9.8$ m/s ² , and we use $g = 10$ m/s ² in calculations. Differences are within reaction-time error.	Gravity pulls on the mkokoteni all the time, so it keeps getting faster, and free fall shows what gravity does on a vertical path: every second adds about 10 m/s to the speed of any object. The cart on the escarpment is not in free fall, so its acceleration is smaller than g (to be investigated in Lesson 5), but the same equations apply to predict its speed and distance.
Lesson 5 Freely Falling Bodies: Explain and Calculate	What did we observe and calculate today? (Include: the mango from the 5 m branch takes $t = 1$ s and lands at $v = 10$ m/s; an object hitting the ground at 30 m/s was dropped from 45 m and fell for 3 s; the cart at $s = 25$ m has $v = 10$ m/s after $t = 5$ s; a 20 m free fall takes 2 s and gives 20 m/s while the cart takes 10 s for the same 20 m/s.)	What did we learn? (Include: with $u = 0$ and $a = g = 10$ m/s ² , the equations of linear motion predict time, speed and height for freely falling bodies; mass does not change the acceleration; the mkokoteni is not in free fall because on the slope its acceleration is $g \times 0.2 = 2$ m/s ² ; free fall is the extreme case of a vertical slope.)	How can we now explain the phenomenon? (Include: the cart keeps getting faster because gravity pulls it down the slope all the time, so its velocity rises by 2 m/s every second; it starts slowly because velocity begins at zero and feels sudden because distance grows as t^2 ; the same equations predict 20 m/s (72 km/h) after 10 s and 100 m, and show why steep roads are dangerous.)
Lesson 6 Road Safety: Stopping Distance and Real-Life Applications of Linear Motion	We observed that a matatu on Mombasa Road needs a longer total stopping distance as speed rises: on a dry road 20 m at 10 m/s (36 km/h), 60 m at 20 m/s (72 km/h) and 120 m at 30 m/s (108 km/h). The braking distances (10 m, 40 m, 90 m dry; 20 m, 80 m, 180 m wet) grew much faster than the speed.	We learned that stopping distance = reaction distance + braking distance. Reaction distance is speed $\times$ reaction time (1.0 s), while braking distance comes from $v^2 = u^2 + 2as$ with $v = 0$, which gives $u^2/2 a $. So doubling the speed multiplies braking distance by 4 and tripling it multiplies it by 9, and a wet road ($a = 2.5$ m/s ²) doubles the braking distance.	We can explain that the runaway mkokoteni reaches 20 m/s (72 km/h) with no brakes and hits the lorry, while a matatu travelling at the same speed on a dry road needs 60 m to stop. Because braking distance depends on the square of speed, steep roads such as the Kinangop descent need warning signs and slow, early braking, and NTSA rules on speed and safe distance save lives. The same equations of linear motion that predict the cart's speed and distance also predict how far a vehicle needs to stop.
Lesson 7 Group Presentations and Building the Class Model: The Mkokoteni's Velocity-Time Graph	What did we observe about the mkokoteni's motion from the table and graph? (For example: velocity rises by 2 m/s every second, giving a straight line through the origin on the v - t graph; distance grows as t^2 , with $s = 25$ m at 5 s but 100 m at 10 s; the	What did we learn today? (For example: the gradient of a velocity-time graph is acceleration (2 m/s ²); the area under it is displacement ($\frac{1}{2} \times 10 \times 20 = 100$ m); a constant acceleration gives a straight line; the s - t graph is a curve; free fall is the same	How do we now explain the driving question? (For example: gravity pulls the mkokoteni down the slope all the time, giving a constant acceleration of $a = g \times 0.2 = 2.0$ m/s ² , so its velocity keeps rising from 0 to 20 m/s in 10 s; it starts slowly because

	cart hits the lorry at 20 m/s, or 72 km/h.) What did you observe in the group presentations about linear motion in real life?	physics with $a = g = 10 \text{ m/s}^2$; stopping distance rises sharply with speed.)	it starts from rest and feels sudden because distance grows as t^2 ; the equations of linear motion and the v-t graph let us predict speed and distance at any moment, and they show why higher speeds need much longer stopping distances.)
Lesson 8 Consolidation, DQB Review and Final Explanation: Why the Mkokoteni Keeps Getting Faster and How We Predict It	We observed that the mkokoteni released from rest on the Limuru-Nairobi escarpment slope speeds up steadily: 0 m/s at 0 s, 4 m/s at 2 s, 10 m/s at 5 s and 20 m/s (72 km/h) at 10 s after 100 m. We observed that a mango dropped from a 5 m branch reaches 10 m/s after 1 s, and that objects dropped from 2.0 m took about 0.64 s regardless of what they were.	We learned that velocity increases by 2 m/s every second because the cart has a constant acceleration of 2.0 m/s^2 ($g \times 0.2$), caused by gravity pulling it down the slope all the time. We learned the four equations of linear motion, how to choose the right one by listing knowns and the unknown, and that on a velocity-time graph the gradient is the acceleration and the area is the displacement. We learned that free fall is the same motion with $a = g = 10 \text{ m/s}^2$ on a vertical path.	The mkokoteni keeps getting faster even with no one pushing because gravity acts on it down the slope all the time, giving a constant acceleration of 2.0 m/s^2 . It starts slowly because its velocity begins at 0 m/s, and the crash feels sudden because the distance grows as t^2 (25 m by 5 s but 100 m by 10 s). Using $v = u + at$, $s = ut + \frac{1}{2}at^2$ and $v^2 = u^2 + 2as$ we can predict that it reaches 20 m/s (72 km/h) after 100 m and 10 s. In a 20 m free fall the same 20 m/s is reached in just 2 s, because the slope gives the cart only 0.2 of g. This is why high speeds need much longer stopping distances and why road-safety rules on steep roads matter.

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